

Draft: On the computation of bundles for fast local convergence in nonsmooth optimization via the Goldstein subdifferential

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Abstract

This is a draft that contains rough ideas for an article that shows how the bundles required for the superlinear steps in the method of [1] can be computed in a more efficient way when the objective is a finite max-type function. It is not meant as a standalone work and should be read in the context of [1]. In particular, several concepts and notations from [1] will be used without proper introduction. Furthermore, results about the Goldstein ε -subdifferential $\partial_\varepsilon f(x)$ will be used, which can be found in [2]. **The theoretical results are in an early state, were not peer-reviewed and may contain errors.**

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1 Introduction

In [1], a higher-order trust-region bundle method was derived that achieves superlinear convergence of serious steps for lower- C^2 functions with polynomial growth. At the heart of the method is Lem. 3.2(b), which shows that for a “sufficiently large” set of bundle points W inside the trust region $\bar{B}_\varepsilon(x)$, the solution of the trust-region subproblem (3.4) has a distance from the minimum x^* that decreases superlinearly w.r.t. the trust-region radius ε . To compute such a set $W \subseteq \bar{B}_\varepsilon(x)$ in practice, Alg. 4.1 was used. While Alg. 4.1 is relatively efficient in terms of oracle calls, it suffers from the fact that the trust-region subproblem (3.4) has to be solved in each of its iterations, which is costly in terms of runtime.

This document is a draft for an article that yields an alternative way of computing the set W in the context of [1]. The idea is to use part (a) of Lem. 3.2 instead of part (b) to perform superlinear steps towards the minimum. Part (a) essentially requires $s(W)$ to contain all “relevant” selection functions in $\bar{B}_\varepsilon(x)$. Since we are in a black-box setting without access to the individual selection functions, it is not possible to work with this requirement directly. (That is why (b) was used for deriving the method in [1].) However, for well-behaved finite max-type functions, it was shown in [3], Thm. 52, that a set $W \subseteq \mathbb{R}^n$ with $s(W) = S$ and $|W| = |S|$ can be computed by a simple gradient sampling method. In Section 2 we show that part of this results can be generalized, by showing that for any finite max-type function and any x close enough to a critical point x^* , any $\xi \in \partial_\varepsilon f(x)$ with ε and $\|\xi\|$ small enough must be a convex combination of subgradients that correspond to a subset $S' \subseteq S$ of selection functions for which x^* is a critical point, i.e., $0 \in \text{conv}(\{\nabla f_s(x^*) : s \in S'\})$. If W' is the set of sample points corresponding to these (sub)gradients, this means that x^* is at least a critical point of the function $F_{s(W)} := \max_{s \in s(W)} f_s$, and we have a sufficiently large bundle W' for $F_{s(W)}$. Unfortunately, we cannot simply apply Lem. 3.2(a) to $F_{s(W)}$ in this case, since we cannot guarantee that the ε -ball used for $\partial_\varepsilon f(x)$ contains x^* , which is crucial for the error estimate in Lem. 3.2. However, in Section 3, we show that the requirements on ε in Lem. 3.2(a) can be significantly weakened, allowing us to perform superlinear steps using W' .

The rest of this draft only contains the above theoretical results, without the explanations and discussion they require.

2 Bundles via elements of $\partial_\varepsilon f(x)$

Assumption (A1). *There are $\mathcal{C}^1$ functions $f_s : \mathbb{R}^n \rightarrow \mathbb{R}$, $s \in S := \{1, \dots, k\}$, such that*

$$f(x) = \max_{i \in S} f_i(x) \quad \forall x \in \mathbb{R}^n.$$

For a finite set Θ denote

$$\Delta_\Theta := \left\{ \lambda = (\lambda_\theta)_\theta \in \mathbb{R}^{|\Theta|} : \lambda_\theta \geq 0 \ \forall \theta \in \Theta, \sum_{\theta \in \Theta} \lambda_\theta = 1 \right\}.$$

It is possible to show that for functions satisfying (A1), $\xi \in \partial_\varepsilon f(x) := \text{conv}(\cup_{y \in \bar{B}_\varepsilon(x)} \partial f(y))$ is equivalent to the existence of a finite set $\Theta = \{(y, s) : y \in \bar{B}_\varepsilon(x), s \in A(y)\} \subseteq \bar{B}_\varepsilon(x) \times S$ and coefficients $\lambda \in \Delta_\Theta$ with $\lambda_{(y,s)} > 0$ for all $(y, s) \in \Theta$ such that

$$\xi = \sum_{(y,s) \in \Theta} \lambda_{(y,s)} \nabla f_s(y). \tag{1}$$

Definition 1. For $x \in \mathbb{R}^n$, $\varepsilon > 0$, and $\xi \in \partial_\varepsilon f(x)$, let

$$\mathcal{S}(\xi) := \{\cup_{(y,s) \in \Theta} \{s\} : \Theta \text{ as in (1)}\}.$$

Theorem 1. Assume that f satisfies (A1). Let $x^* \in \mathbb{R}^n$. Then there is an open neighborhood $U \subseteq \mathbb{R}^n$ of x^* and $\bar{\varepsilon}, \bar{\delta} > 0$ such that if $x \in U$ and $\xi \in \partial_\varepsilon f(x)$ with $\varepsilon \leq \bar{\varepsilon}$ and $\|\xi\| \leq \bar{\delta}$, then for each $S' \in \mathcal{S}(\xi)$, x^* is a critical point of $F_{S'}$.

Proof Part 1: Assume that this does not hold. Then there must be sequences $(x^j)_j \subseteq \mathbb{R}^n$, $(\xi^j)_j \subseteq \mathbb{R}^n$, and $(\varepsilon_j)_j \subseteq \mathbb{R}^{>0}$ with $x^j \rightarrow x^*$, $\varepsilon_j \rightarrow 0$, $\xi^j \in \partial_{\varepsilon_j} f(x^j)$, and $\xi^j \rightarrow 0$, such that for each $j \in \mathbb{N}$, there is some $S' \in \mathcal{S}(\xi^j)$ for which x^* is not a critical point of $F_{S'}$. Since S is finite, we can assume w.l.o.g. that S' is constant for all $j \in \mathbb{N}$.

Part 2: For each $j \in \mathbb{N}$, by definition of $\mathcal{S}(\xi^j)$, there are $\Theta_j \subseteq \bar{B}_{\varepsilon_j}(x^j) \times S$ and $\lambda^j \in \Delta_{\Theta_j}$ with $\{s : (y, s) \in \Theta_j\} = S'$ and

$$\begin{aligned} \xi^j &= \sum_{(y,s) \in \Theta_j} \lambda_{(y,s)} \nabla f_s(y) = \sum_{s' \in S'} \sum_{\substack{(y,s) \in \Theta_j, \\ s=s'}} \lambda_{(y,s)} \nabla f_s(y) \\ &= \sum_{s' \in S'} \mu_{s'}^j \underbrace{\sum_{\substack{(y,s) \in \Theta_j, \\ s=s'}} \frac{\lambda_{(y,s)}}{\mu_{s'}^j} \nabla f_s(y)}_{=: \psi_{s'}^j} \end{aligned} \quad (2)$$

for all $j \in \mathbb{N}$, where $\mu_{s'}^j := \sum_{(y,s) \in \Theta_j, s=s'} \lambda_{(y,s)}$. By construction, for all $s \in S'$ we have $\psi_s^j \in \partial_{\varepsilon_j} f_s(x^j) = \text{conv}(\nabla f_s(\bar{B}_{\varepsilon_j}(x^j)))$ for all $j \in \mathbb{N}$. Since ∇f_s is assumed to be continuous, this yields $\lim_{j \rightarrow \infty} \psi_s^j = \nabla f_s(x^*)$ for all $s \in S'$.

Part 3: By construction it holds that $\mu^j := (\mu_s^j)_{s \in S'} \in \Delta_{S'}$. By compactness, we can assume w.l.o.g. that $(\mu^j)_j$ has a limit $\mu' \in \Delta_{S'}$. Thus, taking the limit $j \rightarrow \infty$ in (2) yields

$$0 = \sum_{i \in S'} \mu'_i \nabla f_i(x^*).$$

Since $s \in A(y)$ for all $(y, s) \in \Theta_j$ we have $S' \subseteq A(x^*)$. This shows that x^* is a critical point of $F_{S'}$, which is a contradiction. $\square$

3 Improved error estimate for higher-order cutting-plane models

Define $\varepsilon(W) := \max_{y \in W} \|y - x^*\|$.

Lemma 1. Assume that $f : U \rightarrow \mathbb{R}$ satisfies (A2) from [1] for $q, p \in \mathbb{N}$, $q \geq p$. Then there are $\varepsilon_{\max} > 0$ with $\bar{B}_{2\varepsilon_{\max}}(x^*) \subseteq U$ and $K \geq 0$ such that for every $\varepsilon \in [0, \varepsilon_{\max}]$, $x \in \bar{B}_{\varepsilon/2}(x^*)$, and finite, nonempty set $W \subseteq \bar{B}_\varepsilon(x)$ with $s(W) = S$ and $\varepsilon(W) \leq \varepsilon/2$, it holds

$$\|\bar{z}^{q,W}(x, \varepsilon) - x^*\| \leq \left(\frac{2K}{\beta} \right)^{1/p} \varepsilon(W)^{\frac{q+1}{p}}.$$

Proof Part 1: Let $\varepsilon_{max} > 0$ with $\bar{B}_{2\varepsilon_{max}}(x^*) \subseteq U$ and let K be the corresponding constant from Lem. 3.2. Let $\varepsilon \in [0, \varepsilon_{max}]$, $x \in \bar{B}_{\varepsilon/2}(x^*)$, and let $W \subseteq \bar{B}_{\varepsilon}(x)$ be a finite, nonempty set with $s(W) = S$ and $\varepsilon(W) \leq \varepsilon/2$. Then by Lem. 3.2(a), we have

$$\bar{\varepsilon} := \|\bar{z}^{q,W}(x, \varepsilon) - x^*\| \leq \left(\frac{2K}{\beta}\right)^{1/p} \varepsilon^{\frac{q+1}{p}}. \quad (3)$$

Since $q \geq p$, we can assume that ε_{max} is small enough to have $\bar{\varepsilon} < \varepsilon/2$. Since $(q+1)/p > 1$, we can further assume that ε_{max} is small enough to have

$$\bar{\varepsilon} > \left(\frac{2K}{\beta}\right)^{1/p} \bar{\varepsilon}^{\frac{q+1}{p}}. \quad (4)$$

Part 2: We first prove, by contradiction, that $\bar{\varepsilon} < \varepsilon(W)$. To this end, assume that $\bar{\varepsilon} \geq \varepsilon(W)$. Since $\bar{\varepsilon} < \varepsilon/2$, this implies $W \subseteq \bar{B}_{\varepsilon(W)}(x^*) \subseteq \bar{B}_{\bar{\varepsilon}}(x^*) \subseteq \bar{B}_{\bar{\varepsilon}}(x) \subseteq U$. Since $\bar{z}^{q,W}(x, \varepsilon) \in \bar{B}_{\bar{\varepsilon}}(x^*)$ we have $\bar{z}^{q,W}(x^*, \bar{\varepsilon}) = \bar{z}^{q,W}(x, \varepsilon)$, so by Lem. 3.2(a) (with center x^* and radius $\bar{\varepsilon}$) it holds

$$\bar{\varepsilon} = \|\bar{z}^{q,W}(x, \varepsilon) - x^*\| = \|\bar{z}^{q,W}(x^*, \bar{\varepsilon}) - x^*\| \stackrel{\text{Lem. 3.2(a)}}{\leq} \left(\frac{2K}{\beta}\right)^{1/p} \bar{\varepsilon}^{\frac{q+1}{p}},$$

which is a contradiction to (4).

Part 3: By assumption we have $W \subseteq \bar{B}_{\varepsilon(W)}(x^*) \subseteq \bar{B}_{\varepsilon/2}(x^*) \subseteq \bar{B}_{\varepsilon}(x)$. As $\bar{z}^{q,W}(x, \varepsilon) \in \bar{B}_{\varepsilon(W)}(x^*)$ by Part 2, this implies that $\bar{z}^{q,W}(x, \varepsilon) = \bar{z}^{q,W}(x^*, \varepsilon(W))$. Application of Lem. 3.2(a) with center x^* and radius $\varepsilon(W)$ completes the proof. $\square$

4 Numerical experiments

By Thm. 1, gradient sampling methods can be used to compute bundles $W = \{y : (y, s) \in \Theta\}$ such that the minimum x^* is a critical point of the function $F_{s(W)}$. If (A2) from [1] holds for this function, then by Lem. 1, solving the trust-region problem (3.4) from [1] for this bundle, with a small enough (but fixed) radius ε , yields a point that decreases the distance to x^* superlinearly w.r.t. $\varepsilon(W)$. For a numerical demonstration of this, see <https://github.com/b-gebken/experiments-on-bundling-for-higher-order-cutting-plane-models>.

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References

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